PURCHASE DESCRIPTION

COUGAR ADULT FISH COLLECTION FACILITY EMERGENCY DIESEL GENERATOR PURCHASE

Scope:

The contractor shall provide one (1) new 150kW Diesel Generator Set, complete with all specified components, features, and warranties. The scope includes the delivery of the unit to the designated location, on-site commissioning and testing, and operator training.

Background:

The U.S. Army Corps of Engineers (USACE) requires a new, reliable backup power source for the Cougar Adult Fish Collection Facility, located at the Cougar Dam. The existing infrastructure necessitates the procurement of a new diesel generator set to ensure continuity of operations during primary power failures. This SOW outlines the requirements for the equipment and all associated service.

Justification:

Current propane generator that supports the COU fish facility with emergency backup power is unreliable. Over the past few years, many issues have come up that we aren't able to fix, and Generac is no longer willing to support troubleshooting and repairs due to the remote location and lack of communications at the site. Reliable emergency power is essential to the proper function of the fish facility, ensuring that it stays operational during critical fish migration seasons, and in compliance with the Willamette Valley BiOp.

Requirements:

Purchase of one (1), new 150kW Diesel Generator Set or approved equal, complete with all specified components, features, and warranties. The scope includes the delivery of the unit to the designated location, on-site commissioning and testing, operator training, drawings, and operation and maintenance manual.

Core Performance Requirements:

- The generator set shall be configured for a 480/277 Volt, 3-Phase, 4-wire electrical system.
- The engine shall be EPA Certified for Stationary Emergency use with an Emergency Standby Power (ESP) rating of 150kw.

Physical Dimensions and Interface Compatibility:

- The overall dimensions of the fully assembled generator package shall not exceed 165 inches in length, 45 inches in width, and 99 inches in height to ensure it fits the existing generator footprint.
- The generator's main electrical load connection panel shall be located on the right-hand side of the enclosure when facing the control panel to align with the existing electrical conduit stub-ups embedded in the concrete pad.
- Any offeror must provide dimensional drawings with their bid to prove compliance with these constraints.

Sound-Attenuated Enclosure:

- The generator set shall be housed within a weather-protective, sound-attenuated enclosure. The enclosure shall limit the average sound level to not exceed 79 dBA at a distance of 23 feet (7 meters).
- The enclosure shall be constructed from galvanized steel components. The exterior paint finish shall be a high-grade, corrosion-resistant coating certified to meet or exceed a 1000-hour salt spray test performance per ASTM B117.
- All access doors shall be side-hinged and equipped with lockable, stainless-steel latches and hardware to prevent corrosion and ensure reliable access.
- Provide removable panels for access to components requiring periodic replacement. The enclosure shall be capable of being removed without disassembly of the engine-generator set or removal of components other than exhaust system.
- The engine exhaust system, including a critical-grade silencer, shall be fully enclosed within the unit's roofline for weather protection and personnel safety from hot surfaces.
- For environmental safety and ease of maintenance, the engine oil and coolant drains shall be piped to the exterior of the enclosure and equipped with shut-off valves.
- The interior of the enclosure shall be equipped with lighting sufficient to illuminate the engine and control panel for nighttime service and inspection.
- The enclosure shall be equipped with a 120VAC, 20-amp GFCI protected duplex receptacle for service and maintenance use.

Integrated Fuel Tank System:

- The generator set shall be equipped with a factory-integrated, dual-wall, UL 142 Listed base fuel tank.
- The tank shall have a 402 gallon capacity, with not less than 395 useable gallons ensuring 35 hours run time at 100% load, and have 5 gallon spill containments with overfill protection.
- The dual-wall base fuel tank shall be equipped with an integrated low-level fuel warning and leak detection system.
- The tank shall be equipped with both a direct-reading mechanical fuel level gauge for local, at-a-glance indication, and an electronic fuel level sender for remote monitoring via the SCADA system.
- The tank shall have a float switch to perform the following functions:
    - **I.** Activate the "Low Fuel Level" alarm at 70 percent of the rated tank capacity.
    - **II.** Activate the “overfill fuel level” alarm at 95% of the rated tank capacity.
- The primary (inner) fuel tank shall include a removable manhole cover to allow for internal inspection, manual cleaning, and service over the life of the unit.
- The tank shall be equipped with both normal and emergency venting systems to comply with relevant safety and fire codes (e.g., NFPA 30).

Engine System:

- The engine shall be designed for stationary applications and shall be complete with ancillaries. The engine shall be a standard production model described in the manufacturer's catalog. The engine shall be turbocharged air-to-air aftercooled. The engine shall be four-stroke-cycle and compression-ignition type.
- The engine shall be equipped with a 120VAC Jacket Water Heater to ensure reliable starting and reduce engine wear in all ambient conditions.
- The engine shall be equipped with a closed crankcase ventilation (CCV) system to properly manage and dispose of engine crankcase fumes.
- The engine shall be delivered filled with engine oil and extended-life coolant rated for operation to -34°C (-30°F).

Fuel System:

- The engine shall be provided with an engine driven pump. The pump shall supply fuel at a minimum rate sufficient to provide the amount of fuel required to meet the performance specified.
- A minimum of one full flow fuel filter shall be provided for each engine. The filter shall be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. The filter shall have inlet and outlet connections plainly marked.
- A relief/bypass valve shall be provided to regulate pressure in the fuel supply line, return excess fuel to a return line, and prevent the build-up of excessive pressure in the fuel system.

Electrical System:

- Alternator: The generator's alternator shall utilize a Permanent Magnet Excitation (PME) system.
- Main Circuit Breaker: The generator package shall include a 100% rated, 3-pole, 250-amp main circuit breaker.
- Battery and Charging System: The package shall include an integrated, UL-listed, 10-amp battery charger to maintain the starting batteries at full capacity.

Control and Monitoring System Requirements:

- The generator control panel shall provide a Modbus RS-485 communication interface for integration with an external SCADA system.
- The controller shall make, at a minimum, the following specific data points available for polling by an external SCADA master system: Generator Running Status, Common Fault Status, Engine Speed, Oil Pressure, Coolant Temperature, and Fuel Level.
- Offerors must provide documentation with their bid, such as a Modbus register map, confirming these data points are available.

Battery System:

- The generator set shall be delivered with heavy-duty starting batteries, battery rack, and all necessary connection cables pre-installed.
- A lockable battery disconnect switch shall be provided to allow for safe maintenance and electrical isolation.

Personnel Safety:

- Exposed moving parts, parts that produce high operating temperatures, parts which may be electrically energized, and parts that may be a hazard to operating personnel during normal operation shall be insulated, fully enclosed, guarded, or fitted with other types of safety devices.
- The safety devices shall be installed so that proper operation of the equipment is not impaired.

Certification and Support Requirements:

- Safety and Performance Certification: The complete generator package (engine, alternator, enclosure, fuel tank) shall be factory-assembled, tested, and certified as a single, integrated unit under UL 2200.
- Manufacture Experience: Statement showing that each component manufacturer has a minimum of three years' experience in the manufacture, assembly and sale of components used with stationary diesel-engine generator sets for commercial and industrial use. The engine-generator set manufacturer/assembler has a minimum of three years experience in the manufacture, assembly and sale of stationary diesel engine-generator sets for commercial and industrial use.
- Manufacturer's certification that the flywheel has been statically and dynamically balanced and is capable of being rotated at 125 percent of rated speed without vibration or damage.
- A certification that each engine generator set passed the factory tests and inspections and a list of the test and inspections.
- Seismic Certification: The generator set shall be mounted on a steel base with spring-type seismic vibration isolators designed to meet IBC certification requirements.
- Extended Service Coverage: The generator package shall include a 5-year / 2500-hour comprehensive extended service coverage plan. The entity responsible for fulfilling the warranty obligations for all major components (engine, alternator, controller) shall be the original engine manufacturer, not a third-party reseller or insurance company.
- On-Site Commissioning: The on-site startup, commissioning, and testing of the generator set shall be performed by a technician factory-certified by the original engine manufacturer to ensure proper installation and validation of the warranty.

Detailed Drawings:

- The contractor shall submit detailed drawings showing the following:
  - **I.** Starting system.
  - **II.** Fuel system.
  - **III.** Cooling system
  - **IV.** Exhaust system.
  - **V.** Lubrication system, including piping, strainers, and filters.
  - **VI.** Location, type, and description of vibration isolating devices.
  - **VII.** Safety system, including wiring schematics.
  - **VIII.** One line schematic and wiring diagrams of the generator, exciter, regulator, governor, and all instrumentation.
  - **IX.** Panel layouts.
  - **X.** Mounting and support for each panel and major piece of electrical equipment.

Operation and Maintenance Data:

- The manufacturers' standard operation and maintenance manual shall include:
  - **I.** Step-by-step procedures for system startup, operation, and shutdown.
  - **II.** Drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems with their controls, alarms, and safety systems.
- The manufacturer's standard maintenance manual shall include:
  - **I.** Procedures for each routine maintenance item, procedures for troubleshooting, factory service, take down overhaul, and repair service manuals with parts list.
  - **II.** Manufacturers recommended maintenance schedule.
  - **III.** A list of spare parts for each piece of equipment and a complete list of materials and supplies needed for operation.

Specials Tools and Filters:

- Two sets of special tools and two sets of filters for required maintenance shall be provided. Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts.
- If the engine generator requires metric tools to perform maintenance and repairs, a set shall be provided.

Engine Diagnostic Software and Hardware:

- The offeror shall provide one (1) complete engine diagnostic software package. The package shall include a lifetime license for the software, the required hardware data link adapter, and all necessary cables to communicate between a laptop computer and the engine's control module.
- The diagnostic software provided shall meet the following mandatory compatibility requirements:
  - **I.** It shall be fully compatible with the offered 150kW generator set's engine and control system, allowing a technician to view real-time operating parameters, view and clear active and logged diagnostic fault codes, and perform engine diagnostic tests.
  - **II.** It shall be fully compatible with the Government's existing fleet of electronically controlled diesel engines.

Submittal Requirements:

- Offerors shall provide technical data sheets, dimensional drawings, and other supporting documentation sufficient to demonstrate compliance with all salient characteristics listed above.

Shipping: Generator shall be delivered to Cougar Adult Fish Collection Facility, Forest Service Rd. 19-410. The Government shall not be responsible for the cost of shipping to the Cougar Location. Hours of Delivery are MON-THU 0800-1600, excluding federal Holidays.

Date and shipping to be coordinated with Jake Moore to ensure crane support is on-site to unload the generator. Contact information below:

Jake Moore
South Willamette Valley Mechanical Working Foreman
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